

문제1

$$F(A,B,C)=AB+A'B+(BC)'+AB$$

중복항 제거 $F=AB+A'B+(BC)'$

또한 $AB+A'B=B$

따라서 $F=B+(BC)'F=B+(BC)'$

DeMorgan 적용 $(B C) '=B'+C'$

최종식 $F=B+B'+C'=1$

진리표

AECF

0001

0011

0101

0111

1001

1011

1101

1111

문제2

$$F(A,B,C,D)=A'BC+ABC+ABD'+A'BD'+CD$$

앞 두 항 묶기 $A'BC+ABC=BC$

다음 두 항 묶기 $ABD'+A'BD'=BD'$

따라서 $F=BC+BD'+CD$

$$F=B(C+D')+CD$$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

문제3

$$F(A,B,C)=(A+B')C+AB+AC'$$

$$F=AC+B'C+AB+AC'$$

$$AC+AC'=A$$

$$F=A+AB+B'C$$

$$F=A+B'C$$

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

문제4

$$F(A,B,C)=AB+A'B+ABC+AB'$$

$$F=AB+A'B+AB'$$

$$F=B+AB'$$

$$B+AB'=A+B$$

$$F=A+B$$

문제5

$$F(A,B,C)=A(B+C)+AB'+AC$$

$$F=AB+AC+AB'+AC$$

$$F=AB+AB'+AC$$

$$F=A+AC$$

$$F=A$$

문제6

$$F(A,B,C,D)=AB+A'C+BC+ABD$$

$$F=AB+A'C+BC$$

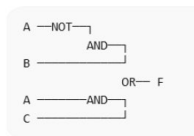
$$F=B(A+C)+A'C$$

$$\text{답1. } F=AB+A'C$$

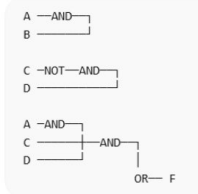
$$A'C+BC=C(A'+B) \text{를 사용하면}$$

$$\text{답2. } F=BC+A'C$$

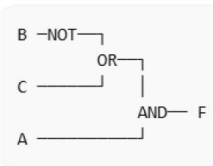
문제7



문제8



문제9



문제 10

1) $F = (AB)'C + B$

2) $F = B + C$

문제 11

1) $F = A[(A+B)' + C]$

2) $F = AC$

문제 12

1) $F = AB + (CD)'$

2) $F = AB + C' + D'$

문제 13

1) $F = (A+C)(B+D)'$

2) $F = AB'D' + CB'D'$

문제 14

1) $F = A[(AB)' + C]$

2) $F = AB' + AC$

문제 15

1) $F = (AB)'(C+D)'$

2) $F = A'C'D' + B'C'D'$